

# Inverse Problems in Geophysics

## Report 2: Transient EM Inversion

### 2. MGPY+MGIN

*Thomas Günther*

*[thomas.guenther@geophysik.tu-freiberg.de](mailto:thomas.guenther@geophysik.tu-freiberg.de)*

# Tasks

1. Define a four-layer model with conductors as 2nd and last layer
2. Generate synthetic data using the given function (using `empymod`)
3. Choose an appropriate time gate vector looking at the apparent resistivity transformation
4. Add Gaussian noise with a relative error of  $\epsilon=3\%$
5. Perform inversion of synthetic data with Marquardt & Occam method
6. Discuss & compare results and all generated plots shortly

Help yourself with [course script](#) and [VES example](#)

# General for both approaches

- use logarithmic transform for both data ( $d = \log \dot{B}$ ) & model ( $m = \log \dot{\rho}$ ) vectors
  - data error weighting by relative error  $\mathbf{d} \rightarrow \log \dot{B}/\epsilon$
  - computation of Jacobian matrix (change model by fraction)
- starting model: homogeneous resistivity of mean  $\rho_a$
- display Jacobian matrix & interpret (Occam: use for designing layers)
- convergence criteria: maximum iteration number, decrease of  $\Phi$ , reaching target  $\chi^2 = 1$  (mean squared error-weighted misfit)
- interpretation of resulting models

# Levenberg-Marquardt algorithm

$$\mathbf{m} = [\log d_1, \log d_2, \log d_2, \log \rho_1, \log \rho_2, \log \rho_3, \log \rho_4]$$

local damping of  $\Delta\mathbf{m}$  (for stabilizing only)

1. Compute forward response & Jacobian matrix (brute force)
2. Solve for model update with some  $\lambda$

$$\Delta\mathbf{m} = (\mathbf{J}^T\mathbf{J} + \lambda\mathbf{I})^{-1}\mathbf{J}^T(\mathbf{d} - \mathbf{f}(\mathbf{m}))$$

3. Lower  $\lambda$  by factor, update model, check convergence & repeat
4. compute & show data information matrix and model covariance matrix

# Smoothness constraints (Occam)

1. Create model layers with increasing thickness & derivative matrix  $\mathbf{C}$
2. Choose starting model  $\mathbf{m}^0$  ( $n=0$ ) & compute model response  $\mathbf{f}(\mathbf{m}^n)$
3. Solve linearized subproblem

$$\Delta\mathbf{m} = (\mathbf{J}^T\mathbf{J} + \lambda\mathbf{C}^T\mathbf{C})^{-1}(\mathbf{J}^T(\mathbf{d} - \mathbf{f}(\mathbf{m})) - \lambda\mathbf{C}^T\mathbf{C}\mathbf{m})$$

4. Update model:  $\mathbf{m}^{n+1} = \mathbf{m}^n + \Delta\mathbf{m}$  ( $\rho^{n+1} = \rho^n e^{\Delta\mathbf{m}}$ )
5. Compute model response & check convergence: proceed with 2.
6. adapt regularization strength  $\lambda$  so that data are fitted within noise
7. Compute model resolution matrix, show matrix and diagonal

**some help (no task)**

# Computation of the Jacobian matrix

- perturbation method (brute force)

$$\frac{\partial f_i(\mathbf{m})}{\partial m_j} \approx \frac{f_i(\mathbf{m} + \delta_j \Delta m) - f_i(\mathbf{m})}{\Delta m}$$

with the Dirac vector  $\delta_j = [0, \dots, 0, 1, 0, \dots, 0]^T$

$\Rightarrow$  one full forward computation for every model parameter

- use logarithms for data and model

# Error weighting

Unweighted residual norm (root-mean square RMS)

$$\|\mathbf{d} - \mathbf{f}(\mathbf{m})\| = \sqrt{1/N \sum (d_i - f_i(\mathbf{m}))^2}$$

Weighting by individual error  $\epsilon_i$  (chi-square value):

$$\chi^2 = \frac{1}{N} \sum \left( \frac{d_i - f_i(\mathbf{m})}{\epsilon_i} \right)^2 \rightarrow \min$$

In case of exact error estimates:  $\chi^2 = 1$



# Error weighting

replace  $d_i$  by  $\hat{d}_i = d_i/\epsilon_i$  leads to

$$\Delta \mathbf{m} = (\hat{\mathbf{J}}^T \hat{\mathbf{J}})^{-1} \hat{\mathbf{J}}^T \Delta \hat{\mathbf{d}}$$

with  $\hat{\mathbf{J}} = \text{diag}(1/\epsilon_i) \cdot \mathbf{J}$

Log-data errors are relative errors

$$\delta \log \rho^a = \frac{\partial \log \rho^a}{\partial \rho^a} \delta \rho^a = \frac{1}{\rho^a} \delta \rho^a = \frac{\delta \rho^a}{\rho^a}$$